

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information and communications technology

Software Requirement Specification

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AIMS Project

Subject: ITSS Software Development

Group 18

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Table of contents

1 Introduction.....	2
1.1 Objective.....	2
1.2 Scope.....	2
1.3 Glossary.....	3
1.4 References.....	5
2 Overall Description.....	6
2.1 Survey.....	6
2.2 Overall requirements.....	7
2.3 Business process.....	8
3 Detailed Requirements.....	11
3.1 Use case 1.....	11
3.2 Use case 2.....	16
3.3 Use case 3.....	21
3.4 Use case 4.....	25
3.5 Use case 5.....	29
3.6 Use case 6.....	33
3.7 Use case 7.....	36
4 Supplementary specification.....	39
4.1 Functionality.....	39
4.2 Usability.....	39
4.3 Reliability.....	39
4.4 Performance.....	40
4.5 Supportability.....	40
4.6 Other requirements.....	40

1 Introduction

1.1 Objective

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the **AIMS (An Internet Media Store)** system. The purpose of this document is to provide a clear and complete description of what the system shall do so that analysts, designers, developers, testers, and project stakeholders can use it as a common reference throughout the software development process. The SRS is written based on the capstone project problem statement and the requirement analysis results produced by the team.

This document is intended for the following audiences:

- Lecturer and teaching assistants: to review and evaluate the correctness and completeness of the requirement analysis.
- Project team members: to ensure a common understanding of the system requirements and project scope.
- Designers and developers: to design and implement the system in accordance with the specified requirements.
- Testers: to derive test cases and verify that the implemented software satisfies the stated requirements.

This document focuses on the required behavior of the system and does not describe implementation details or design solutions.

1.2 Scope

AIMS (An Internet Media Store) is an e-commerce software system developed as a web or desktop application for buying and selling physical media products. The current scope of the system includes the management and sale of four product types: books, newspapers, CDs, and DVDs. The system supports activities for different actors, including customers, product managers, and administrators.

For customers, the system allows browsing products, searching and filtering products, viewing product details, adding items to cart, updating cart contents, placing orders, entering delivery information, paying for orders, viewing order information, and canceling orders before approval. The system also calculates delivery fees, generates

invoice information, records transactions, and sends email notifications related to order status and payment.

For product managers, the system supports product and inventory-related operations such as adding, viewing, editing, deleting, or deactivating products according to business rules, manually adjusting stock with recorded reasons, reviewing pending orders, approving or rejecting orders, and querying product history records.

For administrators, the system supports account and role management functions such as creating, viewing, deactivating, blocking, and unblocking user accounts, assigning roles, triggering password reset processes, and logging sensitive administrative actions.

The system is expected to operate continuously with acceptable reliability and performance. It shall support up to 1,000 simultaneous customers, operate continuously for 300 hours without failure, recover within 1 hour after an incident, and provide a maximum response time of 2 seconds under normal conditions or 5 seconds during peak hours.

At the current stage, the system is limited to physical media products and demonstration-level payment integration using PayPal Sandbox and VietQR. Other product categories, payment gateways, and notification methods may be supported in future versions but are outside the current implementation scope.

1.3 Glossary

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
1	AIMS	The software system being specified in this SRS, short for An Internet Media Store .	AIMS storefront	An e-commerce application for physical media products.
2	SRS	Software Requirements Specification; a document that defines the functional and non-functional requirements of a software system.	This SRS describes the requirements of AIMS.	Project document
3	Product Manager	A user role responsible for managing products, stock, and order approval or rejection.	A Product Manager adds a new CD to the catalog.	Internal actor

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
4	Administrator	A user role responsible for managing user accounts, roles, and sensitive account-related actions.	An Administrator blocks a user account.	Internal actor
5	Customer	An external user who browses products, places orders, pays, and views or cancels orders.	A Customer adds a book to the cart.	External actor
6	Physical media product	A tangible media item sold in AIMS, including books, newspapers, CDs, and DVDs.	A paperback novel or a Blu-ray DVD.	Current supported product category
7	Cart	A temporary collection of products selected by a customer for purchase.	The cart contains 2 books and 1 DVD.	Shopping cart
8	Order	A purchase request submitted by a customer for one or more products.	An order includes shipping and payment information.	Business object
9	Pending processing	The order state after successful payment and before approval or rejection by the Product Manager.	A paid order awaits review in pending processing state.	Order status
10	Invoice	A temporary or final record showing purchased items, quantities, prices, VAT, shipping fee, and total payable amount.	The invoice shows total price including VAT.	Customer-facing information
11	VAT	Value-added tax applied to product prices at the rate specified in the problem statement.	Product prices are stored before 10% VAT.	Business rule
12	Deactivated product	A product whose stock is not zero and therefore cannot be deleted; it is marked unavailable for sale instead.	A newspaper with remaining stock is deactivated instead of deleted.	Product status

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
13	Product history	The stored record of product addition, editing, and deletion/deactivation actions.	Product managers can query product history.	Audit-related term
14	PayPal Sandbox	The testing environment of PayPal used in this course for payment and refund integration.	A customer pays through PayPal Sandbox.	Demonstration integration
15	VietQR	The QR-code-based payment method integrated into AIMS for demonstration purposes.	The system displays a VietQR code for payment.	Payment method
16	Refund	The return of the paid amount to the customer's original payment method when an order is canceled or rejected.	A PayPal-paid order is refunded automatically.	For VietQR, refund is manual
17	Least privilege	A security principle in which users are granted only the permissions necessary for their roles.	Administrators assign only necessary roles to users.	Security principle
18	Hashed password	A protected representation of a password stored in the system instead of the plain text password.	The administrator cannot view the user's actual password.	Security requirement

1.4 References

1. [TEMPLATE-EN] SRS v1.2 (Lecturer: NGUYEN Thi Thu Trang)
2. Lab&Assignment03-SRS (Lecturer: NGUYEN Thi Thu Trang)
3. AIMS-ProblemStatement-ver3.1.1 (Lecturer: NGUYEN Thi Thu Trang)

2 Overall Description

2.1 Survey

AIMS provides a complete workflow for managing the lifecycle of physical media products, from catalog management to customer transactions. The system integrates internal operations such as inventory and account management with external services such as payment gateways.

It supports customer interactions such as browsing, searching, filtering, and purchasing products, while also enabling internal users to manage products, orders, and system configurations. The system maintains logs and history records to ensure traceability and support auditing.

List of Actors and Descriptions

Actor	Type	Description
Customer	External	End-users who interact with the system without authentication. They can browse and search for products, manage a shopping cart, provide delivery information, and complete purchases. They may also view or cancel orders through external communication channels.
Product Manager	Internal	Responsible for managing the product catalog and inventory. They can create, update, deactivate, and remove products, review orders, and handle exceptional cases such as manual refunds or stock adjustments.
Administrator	Internal	Manages user accounts and roles in accordance with security policies. They can create and manage accounts, assign roles, and oversee system access while ensuring sensitive information is protected.
VietQR	External System	A payment service provider that enables QR-based transactions and communicates payment results asynchronously to the system.
PayPal	External System	A payment gateway that processes online transactions and supports additional features such as automated refunds through its API.
Email System	External System	An external service used by AIMS to send invoice information, transaction details, order status notifications, and selected account-related notifications to users

2.2 Overall requirements

2.2.1 Customer

- Browse Products:
The customer browses the product catalog displayed by the system, including the initial random product list shown when the application starts. The customer can also search by title or category and filter by price range.
- View Product Detailed Information:
The customer views the full information of a selected product, including general information and product-type-specific details. From this screen, the customer may also add the product to the cart.
- Check Cart:
The customer reviews products currently added to the cart, including quantity, product price, and total price excluding VAT. During this process, the customer may change quantities or remove products from the cart. If stock is insufficient, the system shall notify the customer.
- Place Order:
The customer proceeds from the cart to checkout. The system checks stock availability, requires delivery information, calculates shipping fees, generates invoice information, and then continues to payment.
- Pay Order:
The customer completes payment using the default QR-based payment method or an alternative payment method through the PayPal gateway. After successful payment, the system records the transaction, sets the order to Pending Processing, and sends invoice and transaction information to the customer's email.
- View Order Info:
The customer can later access order details through the link sent by email. The displayed information includes invoice details, shipping information, and payment transaction information.
- Cancel Order:
The customer may cancel an order before it is approved. If cancellation is accepted, the paid amount is refunded to the original payment method according to the refund rules of the payment gateway.

2.2.2 Product Manager

Product Manager must log in to access the management features corresponding to their roles

- **Manage Products:**
The Product Manager manages the lifecycle of physical media products in the catalog. This includes creating new products, updating existing products, deleting products when stock is zero, and changing product status to deactivated when deletion is not allowed. The system also stores product addition, editing, and deletion history, which can be queried by the Product Manager.
- **Adjust Stock:**
The Product Manager manually adjusts stock quantity when necessary, for example when receiving new inventory or removing damaged items. The system requires a reason to be recorded for each manual stock change.
- **Check Order:**
The Product Manager reviews orders in Pending Processing state. The system displays pending orders and their details for inspection.
- **Approve Order / Reject Order:**
After checking a pending order, the Product Manager may approve or reject it. If an order is rejected, the system notifies the customer and handles refund support according to the payment method.

2.2.3 Administrator

- **Manage Users:**
The Administrator manages internal user accounts and roles, and must log in to access the management features corresponding to their roles. This includes creating accounts, viewing account information, deactivating accounts, blocking or unblocking users, assigning roles, and triggering password reset processes. Sensitive actions such as email updates and password resets must be logged, and affected users must be notified.

2.3 Business process

2.3.1 Purchasing Process

- **Step 1:** The Customer uses Browse Products to find suitable items and may use View Product Detailed Information to inspect the details of a selected product.

- Step 2: The Customer adds desired products to the cart and uses Check Cart to review selected items. During this process, the Customer may change quantities or remove products if necessary.
- Step 3: The Customer selects Place Order. The system checks whether stock is sufficient for the requested quantities. If stock is not sufficient, the Customer must update the cart before continuing.
- Step 4: If stock is sufficient, the system requires the Customer to provide delivery information. Each time the Customer enters or changes the delivery address, the system calculates or recalculates the shipping fee and updates the invoice information.
- Step 5: The Customer continues with Pay Order by using the default QR code payment method or the alternative PayPal-based payment method. After successful payment, the system records the transaction, creates the order in Pending Processing state, and sends invoice and transaction information to the Customer's email.
- Step 6: After the order is created, the Customer may later use the email link to View Order Info or perform Cancel Order before the order is approved.

2.3.2 Order Review Process

- Step 1: The Product Manager performs Login to access the management portal.
- Step 2: The Product Manager uses Check Order to retrieve pending orders and review detailed order information.
- Step 3: Based on stock conditions and business constraints, the Product Manager decides to Approve Order or Reject Order.
- Step 4: If the order is approved or rejected, the system sends a notification email to the customer. If the order is rejected, refund handling is performed according to the payment method capabilities.

2.3.3 Product Management Process

- Step 1: The Product Manager performs Login to access product management functions.
- Step 2: The Product Manager uses Manage Products to create new products, update existing products, or delete selected products according to product rules.
- Step 3: If a selected product has stock greater than zero, the system does not delete it. Instead, the product status is changed to Deactivated so it is no longer available for sale.

- Step 4: Whenever stock is manually changed, the Product Manager uses Adjust Stock and provides a reason for the adjustment. The system records this change for traceability.
- Step 5: The system stores product addition, editing, and deletion history, and the Product Manager may query these records when needed.

2.3.4 User Management Process

- Step 1: The Administrator performs Login to access account management functions.
- Step 2: The Administrator uses Manage Users to create accounts, assign roles, deactivate users, or block and unblock accounts.
- Step 3: If password reset or another sensitive account action is required, the Administrator triggers the action and the system logs it and sends notification to the affected user.

3 Detailed Requirements

3.1 Use case 1

Use Case “Place order”

1. Use case code

UC001

2. Brief Description

This use case describes the sequential interaction between the customer and the AIMS software when the customer decides to finalize their shopping cart and place an order for media products.

3. Actors

- Customer

4. Preconditions

- There must be at least one available physical media product (Book, CD, DVD, etc.) currently placed in the customer's active shopping cart.

5. Basic Flow of Events

1. Customer proceeds to place an order.
2. Software checks whether the available stock is sufficient to fulfill the customer's demand.
3. Software asks customers to set up delivery information.
4. Customer enters the delivery information (see Table 2).
5. Software checks the input information.
6. Software calculates and displays the delivery fee (see Table 4).
7. Customer proceeds to pay the order.
8. Software calls the “Pay order”
9. Software calls the “Pay order”
10. Software calls the “Pay order”
11. Customer completes the payment.
12. Software displays transaction information (see Table 5).
13. Software records the payment transaction information and the successfully paid order.

14. Software puts the order in a pending processing state.

15. Software sends invoice and payment transaction information to the customer's email.

6. Alternative flows

Table 1 – Alternative flows of events for UC place order

No	Location	Condition	Action	Resume location
1.	At step 2	If the stock is insufficient	System prompts the customer that the store is out of the product	Use case end
2.	At step 5	If there are any required fields left blank or invalid information	Software asks customers to update the information.	Resumes at Step 4
3.	At Step 6	If customers update the delivery address	Software will recalculate and display the delivery fee.	Resumes at Step 7
4.	At Step 8	If customers adjust the delivery method or the items they wish to purchase	Software recalculates the delivery fees and updates the corresponding invoice.	Resumes at Step 8

5.	At Step 10	If the customer does not wish to pay by QR code	Customer may switch to an alternative payment method using a credit card.	Resumes at Step 10
6.	At any step	If customers go back to any step or exit the software during the ordering process	Software returns to the previous step or exits the process entirely.	Corresponding step or Use case ends

7. Input data

Table 2 – Input data of delivery information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Receiver Name	Full name of the person receiving the package	Yes	Not blank	Not blank
2.	Phone	Contact number	Yes	- 10 digits - May contains separator “.” and/or space and/or “-” in the middle of the phone number (one separator type), interleave with digits - Starts with “0”	0383841363 0914362222
3.	Province	Delivery region or city	Yes	Chose from a drop-down list	Ha Noi
4.	Address	Detailed street and house information	Yes	Not blank	3 Bo De street

5.	Shipping Instructions	Additional notes for the shipper	No		The house which is opposite the church
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Table 3-Input data of payment

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Payment method	Choice between default QR Code (VietQR) or alternative Credit Card (PayPal)	Yes	Valid payment option	Credit Card
2.	Card details	Required if the customer switches to Credit card	Yes	Valid card number, expiry, CVV	1234-5678-9012-3456

8. Output data

Table 4-Output data of invoice screen

No	Data fields	Description	Display format	Example
1.	Title	Title of the media product	Text	"DVD phim mưa đỏ."
2.	Price	Price of product	-Comma for thousands separator -Positive integer - Right alignment	12,500
3.	Quantity	Number of units of product	-Positive number -Right alignment	2
4.	Amount	Total price of the corresponding product	-Comma for thousands separators	25,000

			-Positive integer -Right alignment	
5.	Subtotal	Total amount of all products	-Comma for thousands separators -Positive integer -Right alignment	150,000
6.	Shipping fee	Calculated delivery cost based on weight/location	-Comma for thousands separators -Positive integer -Right alignment	30,000
7	Total	Final price	Comma for thousands separators -Positive integer -Right alignment	180,000

Table 5-Output data of transaction

No	Data fields	Description	Display format	Example
1.	Customer name		Text	Thanh Huu Dat
2.	Phone number		10 digits - May contains separator “.” and/or space and/or “-” in the middle of the phone number (one separator type), interleave with digits - Starts with “0”	0914362222
3.	Province		Text	Ha Noi
4.	Address		Text	3 Bo De street

5.	Total amount		Right alignment Vietnamese currency (VNĐ) Vietnamese locale	150,000
6.	Transaction ID		5 Positive integer	12345
7.	Transaction content		Text	You have a success transaction at 05/10/2023
8.	Transaction date		dd/mm/yy	05/10/2023

9. Postconditions

- The successfully paid order is recorded and is in a pending processing state.
- The payment transaction information is recorded.
- The customer receives an email with the invoice and payment transaction information.
- The quantity in stock of the purchased products is automatically updated.

3.2 Use case 2

Use Case “Pay Order”

1. Use case code

UC002

2. Brief Description

This use case describes the interactions between the Customer, AIMS software, and VietQR when the customer desires to pay for a placed order using a QR code. AIMS software generates a VietQR QR code for the customer to scan and complete the payment via their banking application. VietQR then sends a payment callback to AIMS software to confirm the transaction result. This use case is

included by the use case “**Place Order**”. If the customer does not wish to pay by QR code, they may switch to credit card payment, which triggers the use case “**Pay Order via PayPal**”.

3. Actors

3.1 Customer

The customer is the end-user who scans the QR code using their banking application to complete the payment, or who may choose to switch to credit card payment via PayPal.

3.2 VietQR

VietQR is an external payment service provider. AIMS software calls VietQR’s API to generate the QR code. VietQR then sends a payment callback to AIMS software with the transaction result.

4. Preconditions

- The use case “Place Order” has been initiated and the customer has confirmed the invoice.
- The invoice information — including the list of products in the cart, quantities, product prices, total product price excluding VAT, total product price including VAT, delivery fee (not subject to VAT), and total amount to be paid — has been calculated and temporarily saved by AIMS software.
- AIMS software has valid VietQR API credentials configured.

5. Basic Flow of Events

Step 1. AIMS software calls the VietQR API to generate a QR code, providing the store’s bank account information, the total amount to be paid (from the temporarily saved invoice), and the transfer content corresponding to the order. (See Table B – Input data for QR code generation.)

Step 2. VietQR returns the QR code to AIMS software.

Step 3. AIMS software displays the payment screen to the customer, showing: the QR code; bank name and account number of the receiving account; transfer content; and the full invoice information including product list (title, quantity, unit price, amount per line), total product price excluding VAT, total product price including VAT, delivery fee (not subject to VAT), and total amount to be paid. (See Table A – Output data of payment screen.) The customer may also choose to switch to credit card payment via PayPal at this step.

Step 4. Customer scans the QR code using their banking application and completes the transfer. (Note: In the test/sandbox environment, the customer triggers VietQR’s test callback API to simulate a successful payment transaction.)

Step 5. VietQR sends a payment callback to AIMS software with the transaction result, including: transaction ID, transaction content, transaction datetime, confirmed amount, and status. (See Table C – Input data from VietQR payment callback.)

Step 6. AIMS software verifies that the callback status is “success” and that the confirmed amount matches the total amount on the invoice.

6. Alternative flows

Table 1-Alternative flows of events for UC Pay Order

No	Location	Condition	Action	Resume location
1	At Step 1	If the customer chooses to pay via PayPal (before the system calls VietQR)	AIMS software inserts the use case “Pay Order via PayPal”.	Use case ends
2	At Step 1	If VietQR fails to generate the QR code (e.g., network error, service unavailable)	- AIMS software notifies the customer that the payment service is temporarily unavailable. - The customer may retry or exit the payment process.	Use case ends
3	At Step 3	If the customer wishes to adjust the items in the order or the delivery information	- AIMS software returns the customer to the delivery information / cart screen. - The customer updates the order or delivery information. - AIMS software recalculates the delivery fee and updates the invoice accordingly.	Resumes at Step 8 of UC Place Order
4	At Step 4	If the payment transaction is reported as failed by VietQR callback	- AIMS software notifies the customer that the payment was unsuccessful. - The customer may retry the payment.	At Step 3
5	At any step	If the customer chooses to go	AIMS software returns the	Resumes at Step 8 of UC Place Order

		back or exit during the payment process	customer to the invoice screen. No payment is recorded.	
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7. Input Data

7.1. Input data for QR code generation request (AIMS to VietQR)

Table B – Input data for VietQR QR code generation

No	Data fields	Description	Mandatory	Valid condition	Example
1	bank_id	Bank identifier of the receiving account	Yes	Valid VietQR bank code	970422
2	account_no	Receiving bank account number	Yes	Non-empty numeric string	0123456789
3	amount	Total amount to be paid (VND), equal to invoice total	Yes	Positive integer	123456
4	add_info	Transfer description / transaction content for the order	Yes	Non-empty string, max 25 characters	AIMS Order #00123
5	callback_url	URL where VietQR will send the payment result callback	Yes	Valid HTTPS URL	https://aims.example.com/callback

7.2. Input data from VietQR payment callback (VietQR to AIMS)

Table C – Input data from VietQR payment callback

No	Data fields	Description	Mandatory	Valid condition	Example
1	transaction_id	Unique transaction identifier assigned by VietQR	Yes	Non-empty string	TXN20231005001
2	transaction_content	Transfer description confirmed by VietQR	Yes	Non-empty string	AIMS Order #00123
3	transaction_datetime	Date and time the transaction was completed	Yes	Valid datetime, format dd/mm/yyyy hh:mm:ss	05/10/2023 14:30:00
4	amount	Amount received and confirmed by VietQR	Yes	Positive integer, must match invoice total	123456
5	status	Result status of the payment transaction	Yes	"success" or "failed"	success

8. Output Data

8.1. Output data of payment screen (AIMS to Customer)

Table A – Output data of payment screen

No	Data fields	Description	Display format	Example
1	QR Code Image	QR code image generated by VietQR for the customer to scan	Image rendered on screen	[QR image]
2	Bank name	Name of the receiving bank	String	MB Bank
3	Account number	Receiving bank account number	Numeric string	0123456789
4	Transfer content	Transfer description corresponding to the order	String	AIMS Order #00123
5	Product title	Title of each media product in the order	String	DVD Mat Biec
6	Quantity	Quantity of each product in the order	Positive integer, right-aligned	2
7	Unit price	Price per unit of each product (excluding VAT)	Comma thousands separator, positive integer, right-aligned	1,234,500
8	Amount	Total per product line (unit price × quantity, excl. VAT)	Comma thousands separator, positive integer, right-aligned	1,234,500
9	Total product price (excl. VAT)	Sum of all product amounts before VAT	Comma thousands separator, right-aligned	1,234,500
10	Total product price (incl. VAT)	Total product price with 10% VAT applied	Comma thousands separator, right-aligned	1,234,500
11	Delivery fee	Shipping fee based on weight and delivery location. Not subject to VAT.	Comma thousands separator, right-aligned	36,360
12	Total amount to be paid	Total = total product price (incl. VAT) + delivery fee	Comma thousands separator, right-aligned	1,234,500

9. Postconditions

9.1. Success

- The payment transaction is confirmed as successful.
- AIMS software records the payment transaction information (transaction ID, transaction content, transaction datetime) and the successfully paid order.
- The order is placed in the “pending processing” state, awaiting review by the product manager.

- An email containing the invoice and payment transaction information is sent to the customer.

9.2. Failure / Cancellation

- If payment is unsuccessful or the customer exits during the payment process, no payment record is created and no order is saved.
- Control returns to the use case “Place Order” at the invoice screen, allowing the customer to retry or go back.
- Note: VietQR does not support an automated refund API. Therefore, if an order that was successfully paid via VietQR is subsequently cancelled (by the customer or rejected by the product manager), no automated refund will be triggered. In such cases, the product manager is responsible for issuing the refund manually (e.g., by contacting the customer and transferring funds outside the system). See the use case “**Cancel Order**” for the full cancellation flow.

3.3 Use case 3

Use Case “Pay Order by Credit Card”

1. Use case code

UC003

2. Brief Description

This use case describes the interaction between the Customer and the AIMS Software, and between AIMS Software and PayPal Sandbox when the Customer wishes to pay for their order using a credit card via the PayPal payment gateway. This use case is an alternative payment flow within the Place Order use case (UC002), triggered when the customer chooses credit card as the payment method instead of the default QR code.

3. Actors

3.1 Customer

The end user who browses products, adds items to the cart, and initiates payment.

3.2 AIMS Software

The e-commerce system that orchestrates the payment flow.

3.3 PayPal Sandbox

External payment service that processes credit card transactions and refunds (sandbox/testing environment).

4. Preconditions

1. The Customer has a valid cart with at least one item.
2. All items in the cart have sufficient stock.
3. The Customer has entered and confirmed valid delivery information.
4. The AIMS Software has displayed the invoice screen with temporary order information.
5. The Customer chooses to switch from the default QR code payment to credit card payment.

5. Basic Flow of Events

1. AIMS Software displays the credit card payment form (see Table A).
2. Customer enters credit card information and submits the form.
3. AIMS Software validates the input fields [is valid = true].
4. AIMS Software sends payment request to PayPal (amount, currency, card info, order reference).
5. PayPal processes the credit card payment.
6. PayPal returns payment result to AIMS Software.
7. AIMS Software checks whether the payment is successful [is successful = true].
8. AIMS Software records payment transaction information.
9. AIMS Software saves the successfully paid order (status = Pending Processing).
10. AIMS Software empties the cart.
11. AIMS Software performs the following two actions in parallel:
 - Displays the order confirmation screen.
 - Sends the invoice and transaction information to the Customer's email.
12. AIMS Software displays the transaction information (see Table B).
13. Customer views the order confirmation screen.
14. Customer decides whether to cancel the order [cancel = no] → use case ends normally.

6. Alternative flows

Table N-Alternative flows of events for UC Pay Order by Credit Card

No	Location	Condition	Action	Resume location
1.	At Step 3	Customer enters invalid or incomplete credit card information	AIMS Software highlights invalid or missing fields and notifies the Customer.	Resumes at Step 2
2.	Step 6	PayPal returns a payment failure response [is successful = false]	AIMS Software displays the payment failure reason to the Customer. Customer decides: • [retry = true]: Return to Step 2 to	Step 2 (retry) or Use case ends (cancel)

			re-enter credit card information. • [retry = false]: Customer selects Cancel; use case ends.	
	Steps 1–4 (before payment submission)	Customer cancels or exits the payment process before submitting	- Customer clicks “Cancel” or closes the PayPal interface. - AIMS Software returns the Customer to the invoice / payment-selection screen. - No payment is charged; the order is NOT finalised.	Use case ends
	Step 13 (after order confirmation displayed)	Customer chooses to cancel the order after successful payment [cancel = yes]	- Customer confirms the cancellation request. - AIMS Software sends a full refund request to PayPal; PayPal processes the refund automatically. - AIMS Software updates the order status to “Cancelled” and sends a cancellation and refund notification to the Customer’s email. - Customer receives the cancellation and refund confirmation.	Use case ends

7. Input data

Table A-Input data of Pay Order by Credit Card

No	Data fields	Description	Mandatory	Valid condition	Example
2.	Cardholder Name	Full name as printed on the credit card	Yes	Non-empty string; letters and spaces only	Nguyen Tuan Dat

2.	Card Number	16-digit credit card number	Yes	16 digits numeric	4111111111111111 1
3.	Expiration Date	Month/Year expiry date on the card	Yes	MM/YY format; must be a future date	12/27
4.	CVV / CVC	3 or 4 digit security code on back of card	Yes	3–4 numeric digits	123
5.	Billing Address	Address associated with the credit card account	No	Free text; max 200 characters	Dong Da, Ha Noi

8. Output data

Table B-Output data of Pay Order by Credit Card

No	Data fields	Description	Display format	Example
1.	Customer Name	Full name of the customer who placed the order	Plain text	Nguyen Tuan Dat
2.	Phone Number	Customer's contact phone number	Plain text (10–11 digits)	0385388626
3.	Shipping Address	Delivery address entered by the customer	Plain text	Luong Dinh Cua
4.	Province / City	Destination province or city	Plain text	Ha Noi
5.	Total Amount	Total amount paid including VAT and delivery fee	Currency (VND)	1,200,000 VND

6.	Transaction ID	Unique identifier returned by PayPal for this transaction	Alphanumeric string	PAYID-123456
7.	Transaction Content	Description / note attached to the transaction	Plain text	Mua hang
8.	Transaction Datetime	Date and time the transaction was completed	DD/MM/YYYY HH:MM:SS	15/03/2026 10:30:18

9. Postconditions

- The payment transaction is successfully processed and recorded by the AIMS Software.
- The order is saved with status “Pending Processing”.
- The Customer receives a confirmation email containing invoice details and transaction information.
- The cart is cleared after successful order placement.

3.4 Use case 4

Use Case “Create product”

1. Use case code

UC004

2. Brief Description

This use case describes the interaction between Product Manager and AIMS software when Product Manager wishes to create a new product in the system

3. Actors

3.1. Name of Actor 1

Product Manager

4. Preconditions

- Product Manager has logged in to access the features corresponding to their role

5. Basic Flow of Events

1. Product Manager requests to add a new product.
2. AIMS software displays the product information input form.
3. Product Manager enters the common product information, including title, category, general description, physical attributes, weight, barcode, original value, and current price.
4. Product Manager selects the product type: Book, Newspaper, CD, or DVD.
5. AIMS software displays the corresponding type-specific fields
6. Product Manager enters the required type-specific information and any optional additional information
7. Product Manager submits the product information.
8. AIMS software validates all entered data, including required fields, business rules, value ranges, and data format.
9. AIMS software saves the new product in the database.
10. AIMS software stores the history of product creation.
11. AIMS software displays a successful creation notification.

6. Alternative flows

Table 1 - Alternative flows of events for UC Create product

No	Location	Condition	Action	Resume location
1.	At Step 8	If the input does not comply with the rules or the format is incorrect	- AIMS software notifies the Product Manager which data fields are invalid or missing, such as blank mandatory fields, invalid date format, invalid dimensions or weight, or current price outside the allowed range. - Product Manager corrects the information.	Resumes at step 3 (for common fields) or step 6 (for type-specific fields)
2.	At any step before Step 9	If Product Manager cancels the operation	AIMS software does not create the product and discards the unsaved input.	Use case ends

7. Input data

Table A - Input data of product information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Title	Name of the product	Yes	Not blank	Doraemon Vol. 1
2.	Category / Product type	Product classification	Yes	Must be one of: Book, Newspaper, CD, DVD	Book
3.	General description	General description of the product, such as condition or color	Yes	Not blank	New, yellow cover
4.	Physical attributes	Height, width, length, and weight	Yes	All values are positive numbers	20 cm x 10 cm x 3 cm; 200 g
5.	Barcode	Product barcode	Yes	Not blank	8934567
6.	Original value	Original value before VAT	Yes	Positive number; excludes 10% VAT	100000 VND
7.	Current price	Selling price before VAT	Yes	Must be between 30% and 150% of original value; excludes 10% VAT	90000 VND
8.	Book details	Author(s), cover type, publisher, publication date	Yes if type = Book	Required fields not blank	Author: Smith; Cover type: Paperback; Publisher: Bach Khoa; Publication date: 15/03/2026
9.	Book additional details	Number of pages, language, genre	No	Optional	250 pages; Vietnamese; Comic

10.	Newspaper details	Editor-in-chief , publisher, publication date	Yes if type = Newspaper	Required fields not blank	Editor-in-chief: Nguyen Van A; Publisher: Thanh Nien; Publication date: 15/03/2026
11.	Newspaper additional details	Issue number, publication frequency, ISSN, language, sections	No	Optional	Weekly; ISSN 1234-5678
12.	CD details	Artists, record label, track list, genre	Yes if type = CD	Required fields not blank; each track must have title and length	Artist: MCK; Record label: RPT; Track: 99%, 3m20s; Genre: Rap
13.	CD additional details	Release date	No	Optional	20/05/2025
14.	DVD details	Disc type, director, runtime, studio, language, subtitles	Yes if type = DVD	Required fields not blank	Disc type: Blu-ray; Director: Nguyen A; Runtime: 120 min
15.	DVD additional details	Release date, genre	No	Optional	15/03/2024; Action

8. Output data

Table B - Output data of product creation

No	Data fields	Description	Display format	Example
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1	Success confirmation	Notification that the product was created successfully	Text	"Product has been created successfully."
2	Invalid data notification	Notification indicating missing or invalid fields or violated business rules	Text	"Publication date format is incorrect."
3	Created product information	Saved product information for confirmation	Text / detail view	Title, type, barcode, current price

9. Postconditions

- A new product is successfully created and saved in the system.
- The history of product creation is stored and can later be queried by the Product Manager.
- If the operation is cancelled, no product is created and the system state remains unchanged.

3.5 Use case 5

Use Case “Update product”

1. Use case code

UC005

2. Brief Description

This use case describes the interaction between Product Manager and AIMS Software when the Product Manager wishes to update the information of an existing product.

3. Actors

3.1 Name of Actor 1

Product Manager

4. Preconditions

- Product Manager has logged in successfully
- The selected product already exists in the system

5. Basic Flow of Events

1. Product Manager requests to update a product
2. AIMS software displays the list of products
3. Product Manager selects one product to update
4. AIMS software displays the current product information
5. Product Manager updates the product information
6. Product Manager submits the updated information
7. AIMS software validates the updated data, including required fields, business rules, value ranges, and format
8. AIMS software updates the product information in the database.
9. AIMS software stores the history of product update.
10. AIMS software displays a successful update notification.

6. Alternative flows

Table 2 - Alternative flows of events for UC Update Product

No	Location	Condition	Action	Resume location
1	At Step 3	If the Product Manager attempts to edit more than one product at a time	AIMS software rejects the action and allows updating only one product at a time.	Resume at Step 2
1.	At Step 7	Updated information is invalid, incomplete, or wrongly formatted	AIMS software notifies the Product Manager about the invalid fields or violated rules.	Resume at Step 5
3	At any step before Step 8	Product Manager cancels the operation	AIMS software does not update the product.	Use case ends

7. Input data

Table C - Input data of product update

No	Data fields	Description	Mandatory	Valid condition	Example
1	Product ID	ID of selected product	Yes	Must exist in the system	P001

2	Title	Name of the product	Yes	Not blank	Doraemon Vol. 1
3	Category / Product type	Product classification	Yes	Must be Book, Newspaper, CD, or DVD	Book
4	General description	General description of the product	Yes	Not blank	New edition
5	Physical attributes	Height, width, length, and weight	Yes	Positive numbers	20 cm x 10 cm x 3 cm; 220 g
6	Barcode	Product barcode	Yes	Not blank	8934567
7	Original value	Original value before VAT	Yes	Positive number	100000 VND
8	Current price	Selling price before VAT	Yes	Between 30% and 150% of original value; excludes 10% VAT	95000 VND
9	Book details	Author(s), cover type, publisher, publication date	Yes if type = Book	Required fields not blank	Smith; Hardcover; Bach Khoa; 15/03/2026
10	Book additional details	Number of pages, language, genre	No	Optional	250; English; Comic
11	Newspaper details	Editor-in-chief, publisher, publication date	Yes if type = Newspaper	Required fields not blank	Nguyen Van A; Thanh Nien; 15/03/2026

12	Newspaper additional details	Issue number, publication frequency, ISSN, language, sections	No	Optional	No. 25; Weekly
13	CD details	Artists, record label, track list, genre	Yes if type = CD	Required fields not blank; each track has title and length	MCK; RPT; Track 1: 99% - 3m20s
14	CD additional details	Release date	No	Optional	20/05/2025
15	DVD details	Disc type, director, runtime, studio, language, subtitles	Yes if type = DVD	Required fields not blank	Blu-ray; Nguyen A; 120 min
16	DVD additional details	Release date, genre	No	Optional	15/03/2024; Action

8. Output data

Table D - Output data of product update

No	Data fields	Description	Display format	Example
1	Success confirmation	Notification that the product was updated successfully	Text	"Product has been updated successfully."
2	Invalid data notification	Notification indicating invalid fields or violated business rules	Text	"Current price must be between 30% and 150% of original value."
3	Updated product information	Updated product information for confirmation	Text / detail view	Updated title, price, publication date

4	Update history record	Confirmation that the update action has been logged in product history	Text / internal record	"Update history stored."
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9. Postconditions

- Product information is successfully updated in the system
- The history of product editing is stored and can be queried by the Product Manager
- If the operation is cancelled, no product information is changed

3.6 Use case 6

Use Case “Delete product”

1. Use case code

UC006

2. Brief Description

This use case describes the interaction between Product Manager and AIMS software when the Product Manager wishes(es) to delete products from the system.

3. Actors

3.1 Name of Actor 1

Product Manager

4. Preconditions

- Product Manager has logged in successfully
- The selected product already exists in the system

5. Basic Flow of Events

1. Product Manager requests to delete product(s).

2. AIMS Software displays the product list.
3. Product Manager selects one or more products to delete.
4. AIMS Software checks that the number of selected products does not exceed 10 products per request.
5. AIMS Software checks that the Product Manager has not exceeded the limit of 20 deleted products per day.
6. AIMS Software checks the stock quantity of each selected product.
7. AIMS Software deletes all selected products whose stock quantity is equal to 0.
8. AIMS Software stores the deletion history.
9. AIMS Software notifies the Product Manager of the deletion result

6. Alternative flows

Table 3 - Alternative flows of events for UC Delete Product

No	Location	Condition	Action	Resume location
1	At Step 4	If the Product Manager selects more than 10 products in one request	AIMS Software notifies the Product Manager that the request exceeds the allowed limit	Resumes at Step 3
2	At Step 5	If deleting the selected products causes the Product Manager to exceed the limit of 20 deleted products per day	AIMS Software notifies the Product Manager that the daily deletion limit has been exceeded	Use case ends
3	At Step 6	If one or more selected products have stock quantity greater than 0	- AIMS software changes the status of those products to "deactivated" meaning it will no longer be available for sale - AIMS software stores this change in product history and notifies the Product Manager of the result	Resume at Step 8

4	At any step before Step 7	If the Product Manager cancels the delete operation	AIMS Software does not delete or deactivate any product	Use case ends
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7. Input data

Table E - Input data of product deletion

No	Data fields	Description	Mandatory	Valid condition	Example
1	Selected products	List of products chosen for deletion	Yes	Maximum 10 products per request	[P001, P002]
2	Daily deletion count	Number of products already deleted on the current day by the same Product Manager	Yes	Total deleted products per day must not exceed 20	12
3	Stock quantity	Current stock quantity of each selected product	Yes	Product can only be deleted if stock = 0	0

8. Output data

Table F - Output data of product deletion

No	Data fields	Description	Display format	Example
1	Invalid notification	Notification when the delete operation violates business rules	Text	"Cannot delete more than 10 products in one request."
2	Daily limit notification	Notification when daily deletion limit is exceeded	Text	"Cannot delete more than 20 products per day."

3	Product status	Status shown for products that cannot be deleted because stock is not zero	Text	"Deactivated"
4	Deletion result	Result summary of the operation	Text	"3 products deleted, 2 products deactivated."
5	Deletion history record	Confirmation that deletion/deactivation actions were logged	Text / internal record	"Deletion history stored."

9. Postconditions

- Selected products are deleted or deactivated in the database
- The history of product deletion is stored and can be queried by the Product Manager
- If the operation is cancelled or rejected due to rule violations, the system state remains unchanged

3.7 Use case 7

Use Case “View Product Detail”

1. Use case code

UC007

2. Brief Description

This use case describes how a customer views the detailed information of a selected product, including general product information and type-specific information depending on the product type.

3. Actors

Customer

3.1 Name of Actor 1

Product Manager

4. Preconditions

The customer has accessed the AIMS system and the product list has been displayed on the system

5. Basic Flow of Events

1. The actor accesses the product list
2. The customer selects a product to view
3. The system receives the request to view the product detail
4. The system retrieves the product information from the database
5. The system displays the detail information of the product to the customer

6. Alternative flows

Table N-Alternative flows of events for View Product Detail

No	Location	Condition	Action	Resume location
	At Step 4	The selected product does not exist	System display an error message that	Use case ends

			the product is not found	
	At Step 5	The selected product is unavaible	System informs the customer that the product is no longer available for sale	Use case ends or resumes at Step 1/2
	At step 4	A system error occurs while retrieveing product details	System displays an error message	Use case ends

7. Input data

Table A-Input data of View Product Detail

No	Data fields	Description	Mandator y	Valid condition	Example
	Selected product	Product selected by the customer	YES	MUST correspond to an existing product in the system	P888

8. Output data

Table B-Output data of View Product Detail

No	Data fields	Description	Display format	Example
1	Title	Product title	Text	One piece
2	Category	Product category	Text	Book
3	Price	Current price of the product	Currency	888888 VNĐ

9. Postconditions

The detailed information of the selected product is displayed to the customer

4 Supplementary specification

4.1 Functionality

- **Authentication & Access:** Administrators and product managers must log in to access role-specific features. Administrators are responsible for managing user accounts in accordance with the principle of least privilege and data protection standards. Administrators may trigger password reset processes but cannot access users' actual passwords, as all passwords are securely hashed. For customers, no login is required.
- **Auditing & Logging:** The software will store the history of products addition, editing and deletion. Product managers can query these histories whenever they want. Manual stock adjustments must be performed by the product manager, and the reason for the stock change must be explicitly recorded. The software records the payment transaction information and the successfully paid order. Any sensitive administrative actions, such as email updates or password resets, must be logged.
- **Notifications:** Any sensitive administrative actions must be automatically notified to the affected users. The software will notify the product manager if any operation is invalid. The software will send invoice and payment transaction information to the customer's email. Once an order is either accepted or rejected, a notification email will be sent to the customer who placed the order. If an order paid via VietQR is rejected, the system will notify the Product Manager that a manual refund to the customer is required for this payment method.
- **System Constraints:** The price of the product must always be between 30% and 150% of the product's original value to avoid price inflation or undercutting. For security reasons, managers cannot delete more than twenty products per day. A product can only be deleted if its stock is equal to zero, otherwise it cannot be deleted; instead, its status will be changed to "deactivated".

4.2 Usability

- **Ease of Use:** The e-commerce software operates 24/7 allowing new users to easily familiarize themselves.
- **Navigation Flexibility:** Customers can go back to any step or exit the software during the ordering process.

4.3 Reliability

- **Availability:** The software operates 24/7.
- **Fault Tolerance:** This software can operate continuously for 300 hours without failure.

- Recoverability: The software can resume normal operation within a maximum of 1 hour after an incident.

4.4 Performance

- Capacity: This software can serve up to 1,000 customers simultaneously without significantly reducing performance.
- Response Time: The maximum response time of the software is 2 seconds under normal conditions or 5 seconds during peak hours.

4.5 Supportability

- Extensibility for Products: Currently, AIMS supports only the buying and selling of physical media products, including books, newspapers, CDs, and DVDs (other types of products may be supported in the future).
- Extensibility for Services: Other payment gateways may be supported in the future. In the future, additional notification methods such as SMS or push notifications may also be supported.

4.6 Other requirements

- Platform: AIMS is an e-commerce software (web or desktop application).
- Integration: Users will integrate payments using PayPal Sandbox and VietQR (via QR code). Currently, since only PayPal supports the refund API, the system does not handle refunds for VietQR payments. In such cases, the Product Manager is responsible for issuing refunds manually (e.g., contacting the customer and transferring funds outside the system).